A Study of Controllable Pragmatics in Low-Resource Language Generation: Burmese with mT5

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Abstract

Generating contextually appropriate language requires not only semantic correctness but also sensitivity to sociolinguistic factors such as politeness, tone, and power relations. This challenge is particularly pronounced in low-resource languages like Burmese, where pragmatic meaning is deeply embedded in social context.

In this work, we propose a framework for context-aware pragmatic generation using a multilingual text-to-text model. We leverage a structured dataset of Burmese conversational expressions annotated with contextual and sociolinguistic attributes, and transform these attributes into textual prompts for fine-tuning an mT5 model.

We design a controlled experimental setup to analyze the contribution of contextual and pragmatic attributes to generation quality, and to examine how different sociolinguistic factors influence model behavior. In addition to standard text similarity metrics such as BLEU and chrF, we introduce a pragmatic consistency evaluation based on attribute classification to assess whether generated outputs align with intended sociolinguistic properties.

We hypothesize that incorporating contextual and pragmatic attributes significantly improves the appropriateness of generated utterances compared to baseline models. In addition, we aim to identify which pragmatic dimensions are more effectively learned and which remain challenging under low-resource conditions. This work highlights the importance of modeling pragmatics in multilingual generation and provides insights into controllable language generation for low-resource languages.

1 Introduction

Recent advances in large-scale language models have significantly improved the quality of text generation across a wide range of tasks. However, most existing approaches primarily focus on semantic correctness, often overlooking the role of pragmatics—how meaning is shaped by context, social relationships, and communicative intent.

This limitation becomes particularly evident in languages such as Burmese, where linguistic expressions are highly sensitive to sociolinguistic factors including politeness, tone, and power relations between speakers. In such settings, generating grammatically correct text is insufficient; models must also produce contextually appropriate and socially aligned utterances.

Despite growing interest in controllable text generation, most prior work has focused on high-resource languages such as English and on relatively coarse attributes such as sentiment or topic. Furthermore, existing approaches often treat attributes independently and do not explicitly model the interaction between multiple contextual and pragmatic factors. As a result, there remains a gap in developing generation systems that can effectively model fine-grained pragmatics in low-resource languages.

In this work, beyond improving generation performance, we aim to systematically study how different contextual and sociolinguistic attributes influence controllable text generation in a low-resource setting. Specifically, we investigate the relative contribution of semantic, contextual, and pragmatic factors, and analyze which aspects of pragmatics are more readily captured by multilingual models.

To evaluate the effectiveness of our approach, we design a dual evaluation framework that combines traditional text similarity metrics with a pragmatic consistency measure based on attribute classification. This allows us to assess not only the linguistic quality of generated text but also its alignment with intended sociolinguistic properties.

The main contributions of this work are as follows:

- We propose a framework for controlled pragmatic generation in Burmese using a multilingual text-to-text model.
- We introduce a prompt-based formulation that incorporates multiple contextual and sociolinguistic attributes for generation.
- We conduct a controlled study to analyze the contribution of different contextual and pragmatic attributes.
- We design a dual evaluation strategy that measures both text similarity and pragmatic consistency.
- We provide attribute-level and qualitative insights into the challenges of modeling pragmatics in low-resource languages.

2 Research Questions

This study investigates the role of contextual and sociolinguistic information in controlled text generation for a low-resource language. We formulate the following research questions:

RQ1: How does fine-tuning a multilingual text-to-text model affect its ability to generate Burmese utterances compared to a zero-shot baseline?

RQ2: To what extent does incorporating contextual information improve the quality and appropriateness of generated utterances?

RQ3: Which sociolinguistic attributes (e.g., politeness, tone, and power relations) contribute most to generation quality, and which are more challenging for the model to learn in a low-resource setting?

3 Related Work

3.1 Multilingual Text-to-Text Models

Recent advances in multilingual pre-trained models have enabled effective transfer across languages, including low-resource settings. The Text-to-Text Transfer Transformer (T5) framework unifies all NLP tasks into a text-to-text format, allowing flexible conditioning on structured inputs. Building on this, mT5 extends the architecture to a massively multilingual setting, covering over 100 languages and demonstrating strong performance across multilingual benchmarks (Xue et al., 2021).

The text-to-text paradigm is particularly suitable for generation tasks where structured information must be mapped into natural language outputs. In this work, we adopt mT5 as the backbone model due to its ability to handle multilingual input and generate fluent Burmese text, making it appropriate for low-resource pragmatic generation.

3.2 Controlled Text Generation

Controlling specific attributes of generated text, such as sentiment, style, or topic, has been an active area of research. Early approaches relied on supervised fine-tuning or reinforcement learning to guide generation toward desired attributes. More recent work has explored plug-and-play methods that allow control without retraining the entire model.

Recent work has explored plug-and-play approaches for controllable generation. Madotto et al. (2020) propose plug-and-play conversational models that enable control over attributes such as sentiment and topic without requiring task-specific fine-tuning. Their approach demonstrates that attribute control can significantly improve response quality and consistency.

However, these methods introduce substantial computational overhead during inference and require careful tuning of control parameters. In contrast, our approach focuses on efficient fine-tuning of a text-to-text model, enabling attribute-aware generation without expensive decoding-time optimization.

3.3 Context-Aware and Pragmatic Generation

Generating contextually appropriate and socially aware responses remains a key challenge in dialogue systems. Prior work has explored various aspects of context-aware generation, including personalization (Li et al., 2016), and emotional dialogue modeling (Rashkin et al., 2019).

In particular, controllable attributes such as sentiment, tone, and style have been shown to significantly impact human judgments of conversational quality (See et al., 2019). These studies highlight the importance of incorporating pragmatic and sociolinguistic factors into generation models.

Despite these advances, most existing work focuses on high-resource languages such as English, and there is limited research on pragmatic language generation in low-resource languages. Furthermore, prior approaches often treat attributes independently rather than modeling multiple interacting sociolinguistic factors.

In this work, we address these gaps by introducing a framework for Burmese pragmatic generation that jointly models semantic intent and multiple contextual attributes, including politeness, tone,

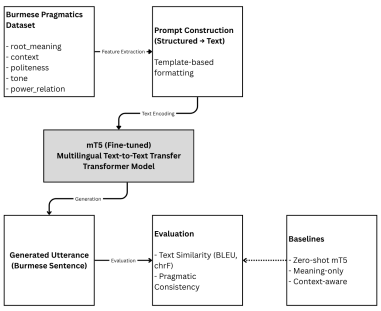

Figure 1: Overview of the proposed framework. Structured inputs are converted into textual prompts and used to fine-tune an mT5 model for generating Burmese utterances. Outputs are evaluated using both text similarity and pragmatic consistency metrics, with baseline models included for comparison.

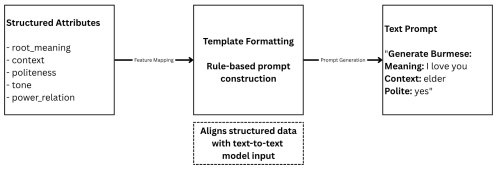

Figure 2: Prompt construction process. Structured attributes are converted into a textual prompt using a template-based approach, enabling compatibility with text-to-text models.

and power relations. By leveraging a multilingual text-to-text model, we enable controlled generation in a low-resource setting while maintaining both linguistic fluency and pragmatic consistency.

4 Methodology

We propose a framework for context-aware pragmatic generation in Burmese using a multilingual text-to-text model. The system takes structured inputs consisting of semantic intent and sociolinguistic attributes, converts them into textual prompts, and generates contextually appropriate utterances. An overview of the proposed framework is shown in Figure 1.

4.1 Dataset

We utilize the Burmese Contextual Pragmatics dataset (freococo, 2024), which consists of approximately 2,200 samples covering 22 core conversational intents. Each sample contains a root semantic meaning along with multiple contextual and sociolinguistic attributes, including context, politeness, tone, and power relations.

The dataset provides diverse realizations of the same semantic intent under different social conditions, enabling controlled experimentation on how contextual and pragmatic factors influence language generation. Each entry includes a corresponding Burmese utterance, which serves as the target output for training.

4.2 Data Preprocessing

We preprocess the dataset by selecting a subset of relevant attributes for controlled generation. Specifically, we retain root meaning, context, politeness, tone, and power relations as input features, and use the corresponding utterance as the target output.

The dataset is split into training, validation, and test sets to ensure fair evaluation. All models are trained on the training set and evaluated on a held-out test set. This setup prevents data leakage and ensures that the evaluation reflects generalization to unseen samples.

4.3 Prompt Construction

To align structured dataset attributes with the input requirements of text-to-text models, we convert each sample into a textual prompt. This process involves formatting the selected attributes into a consistent template that conditions the generation process.

Figure 2 illustrates the prompt construction pipeline.

Each input is transformed into a structured prompt of the following form:

Generate Burmese sentence:
Meaning: <root_meaning>
Context: <context>
Politeness: <politeness>
Tone: <tone>
Power: <power_relation>

This formulation allows the model to condition its output on both semantic intent and sociolinguistic constraints.

4.4 Model Setup

We adopt mT5-small (Xue et al., 2021), a multilingual text-to-text transformer model, as the backbone for generation. The model is pre-trained on a large multilingual corpus and is capable of handling Burmese text.

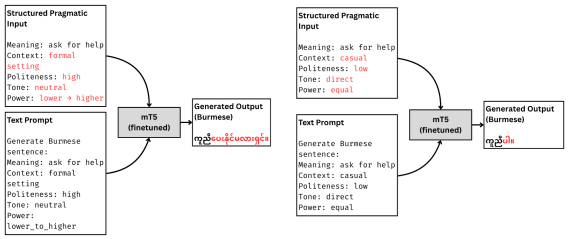

Figure 3: Illustration of controllable pragmatic generation. Different sociolinguistic attributes (e.g., context, politeness, tone, and power relations) result in different generated Burmese outputs for the same underlying semantic intent.

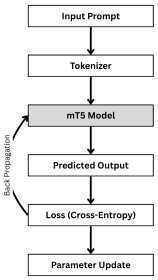

Figure 4: Fine-tuning process. The model takes tokenized input prompts, generates predictions, and updates parameters through backpropagation based on cross-entropy loss.

We fine-tune the model on the constructed prompts and corresponding target utterances, enabling it to learn mappings from structured inputs to contextually appropriate outputs.

4.5 Fine-Tuning Procedure

The fine-tuning process follows a standard sequence-to-sequence training paradigm. Input prompts are tokenized and passed through the mT5 model to generate predicted outputs. The model is trained to minimize cross-entropy loss between the generated output and the ground truth utterance.

Figure 4 illustrates the training process.

4.6 Baselines

To evaluate the contribution of contextual and pragmatic information, we design multiple baseline models with varying input configurations:

- **Zero-shot mT5**: Pretrained model without fine-tuning.

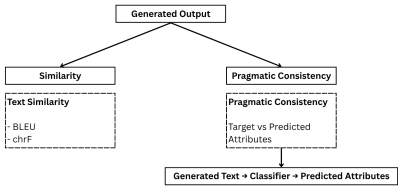

Figure 5: Evaluation framework. Generated outputs are evaluated using text similarity metrics and pragmatic consistency through attribute classification and comparison with target labels.

- **Meaning-only model**: Fine-tuned using only root meaning.
- **Context-aware model**: Fine-tuned using root meaning and context.

These baselines allow us to isolate the effects of fine-tuning, contextual information, and pragmatic attributes.

4.7 Evaluation Strategy

We evaluate the generated outputs using both text similarity metrics and pragmatic consistency measures. The evaluation framework is shown in Figure 5.

Text similarity is measured using BLEU (Papineni et al., 2002) and chrF (Popović, 2015) scores, which quantify the closeness of generated outputs to reference utterances.

To assess pragmatic consistency, we employ an attribute classification approach. Generated utterances are passed through a classifier to predict attributes such as politeness or tone. These predictions serve as a proxy for pragmatic consistency. These predictions are then compared with the target attributes from the dataset to determine whether the generated output adheres to the intended sociolinguistic constraints.

This dual evaluation approach allows us to measure both linguistic accuracy and contextual appropriateness. In addition, we report attribute-level consistency scores to analyze how well each sociolinguistic dimension is captured by the model.

4.8 Analysis Plan

In addition to comparing baseline models, we conduct further analysis to better understand the role of different input components. Specifically, we

perform ablation experiments to assess the contribution of individual attributes such as context, politeness, tone, and power relations.

We also report attribute-level performance by evaluating pragmatic consistency separately for each sociolinguistic dimension. Furthermore, we conduct qualitative error analysis to examine cases where generated outputs fail to align with intended attributes. Where possible, we analyze performance differences across frequent and rare attribute combinations to better understand model behavior under data sparsity.

5 Expected Results

We expect that fine-tuning a multilingual text-to-text model on the Burmese Contextual Pragmatics dataset will significantly improve generation quality compared to a zero-shot baseline. Specifically, the fine-tuned model is anticipated to produce outputs that are more fluent and better aligned with the intended semantic meaning.

In terms of contextual modeling, we expect that incorporating contextual information will improve the appropriateness of generated utterances. Models that include contextual inputs are likely to generate responses that better reflect situational factors, compared to models trained solely on root meaning.

Furthermore, we hypothesize that the inclusion of explicit pragmatic attributes such as politeness, tone, and power relations will improve the model's ability to generate socially appropriate and consistent outputs. In particular, we expect higher pragmatic consistency scores for models conditioned on these attributes, indicating better alignment between intended and generated sociolinguistic features.

However, due to the limited size of the dataset and the restricted number of root meanings, we expect the model's generalization ability to be constrained. The model may struggle to produce appropriate outputs for unseen or more complex semantic inputs beyond the predefined categories.

Overall, we expect a clear performance progression across model variants, with the full model incorporating semantic, contextual, and pragmatic information achieving the best performance in both text similarity and pragmatic consistency metrics.

We also expect that not all pragmatic attributes contribute equally to performance. For example, attributes such as politeness may be easier for the model to learn, while more complex relational factors such as power dynamics may remain challenging. Additionally, we anticipate that improvements in text similarity metrics may not always correlate with gains in pragmatic consistency, particularly in cases involving subtle sociolinguistic distinctions or rare attribute combinations.

6 Discussion

This work explores the role of contextual and sociolinguistic factors in text generation for a low-resource language. Unlike conventional approaches that focus primarily on semantic correctness, our framework emphasizes the importance of pragmatic appropriateness, which is essential for natural language use in Burmese.

By incorporating structured attributes such as politeness, tone, and power relations, the proposed approach aims to enable more controlled and socially aware generation. This is particularly relevant for languages where meaning is highly dependent on social context, and where inappropriate usage may lead to misunderstandings.

The comparison across multiple model configurations allows us to analyze the contribution of different input components. This controlled setup provides insights into how contextual and pragmatic information influence generation quality, beyond what is captured by traditional text similarity metrics.

Furthermore, the inclusion of pragmatic consistency as an evaluation dimension highlights the limitations of relying solely on lexical similarity. This dual evaluation framework reflects a more comprehensive assessment of generation quality, particularly in sociolinguistically rich settings.

Overall, this study contributes to a growing body of work that seeks to improve controllability and contextual awareness in multilingual language models, especially for low-resource languages.

7 Summary of Progress

At the current stage of the project, we have completed dataset exploration and analysis, identifying key attributes relevant for controlled generation. We have also designed the overall experimental framework, including prompt construction, model configurations, and evaluation strategies.

In addition, we have developed a structured methodology and created diagrams illustrating the system pipeline, prompt construction process,

fine-tuning procedure, and evaluation framework. These components provide a clear blueprint for the implementation phase.

The next steps involve implementing the training pipeline, fine-tuning the mT5 model under different input configurations, and conducting systematic evaluations. We will also develop the attribute classifier used for pragmatic consistency assessment.

Potential challenges include limited dataset size, computational constraints, and the difficulty of accurately modeling nuanced sociolinguistic features. Addressing these challenges will be a key focus in the upcoming phases of the project.

Limitations

This study has several limitations. First, the dataset is relatively small and limited to a predefined set of 22 root meanings, which may restrict the model's ability to generalize to broader or more complex language use cases.

Second, the evaluation of pragmatic consistency relies on an attribute classifier trained on the same dataset distribution. While a held-out test set is used to mitigate data leakage, this approach may still introduce bias and may not fully capture real-world pragmatic correctness.

Third, the scope of the study is limited to Burmese, and the findings may not directly generalize to other languages with different sociolinguistic structures.

Finally, due to computational constraints, the experiments are conducted using a smaller variant of the mT5 model, which may limit the overall performance compared to larger-scale models.

8 Conclusion

In this work, we propose a framework for context-aware pragmatic generation in Burmese using a multilingual text-to-text model. By incorporating structured contextual and sociolinguistic attributes, the proposed approach aims to improve the appropriateness and controllability of generated utterances.

We design a controlled experimental setup to evaluate the impact of contextual and pragmatic information, and introduce a dual evaluation strategy that considers both text similarity and pragmatic consistency. Through this study, we aim to provide insights into the challenges and opportunities of modeling pragmatics in low-resource languages.

The proposed framework serves as a foundation for future work on socially aware language generation and highlights the importance of incorporating contextual and cultural factors in multilingual NLP systems.

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